

IIR Based Butterworth Filter Design to Denoise an Audio Signal

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Abstract—Digital filters are mandatory for digital signal processing, with the increased use of communication via digital signal speech data in real world applications, noise reduction in the data is becoming significantly important. Background noises in speech data are coming from different sources including traffic, crowd and uncontrollable environmental parameters. The purpose of this paper is to design and explore digital filter in order to reduce noise in the speech signal. This paper presents digital filter dispelling the unwanted signals or noise from the required signal and enhances better performances of the signal. The extracted features of the digital filter have been analyzed to acquire better output of the signal by using IIR Butterworth filter. It provides different designed parameters of IIR filter to achieve the desired result. MATLAB FDA tool is considered to find out the different responses of a digital filter, about six parameters like- Phase response, Magnitude response, Magnitude & Phase response, Step response, Pole/Zero Plot, Impulse response are used to analyze the filter responses. Some selected audio signals are used for observing the empirical response of high pass, low pass, band stop filter and band pass filter. A special tool is developed for this observation purpose.

Keywords— Digital Filters, Signal Processing, IIR Butterworth Filter, Responses, Impulse, DSP

I. INTRODUCTION

For any type of communication system, signal processing is the centerpiece. To avoid unwanted signals, processing of signal is required. Analog and digital signal processing can be used to process the information. More flexibility and better control of precision is provided by digital signal processing compared to analog signal processing. Filter is applied to signal processing. To process the raw signals, a digital filter is essential. The system which performs operations on a discrete time signal, sampled to reduce or increase certain characteristics of that signal is called a digital signal [1]. Various statistical methods can be introduced to observe the performance of a digital filter. Several observation methods can also be used in the design and often form the origin of a filter element. Basically, we denote filters by calculating how they will react to a simple input such as a pulse. Another code could then wrap that data to subtract the filter response from more aggregated signals. In this study, different filter forms such as high-pass filter, low-pass filter, notch filter and band-pass filter were used to analyze the performance of the digital filter. A selected number of audio signals are used to observe responses of various digital filters and also monitor and play outputs of the filters. The objective of this study was to discover the best performance of digital filters. To achieve this goal, it is necessary to optimize various parameters of digital

filter . Parameters such as amplitude response, phase delay , phase response, impulse response, graph pole/zero and step response are used to optimize the results, and determine the best output. To get the best yield, a Butterworth IIR filter was used. The cooperation between the attenuation and the phase response is known as the Butterworth filter . It is sometimes called a flat filter at most because; it has no flow in the stopband or in the band. The Butterworth filter obtains its flatness at the cost of a relatively large evolution from the bandwidth to the stop band. The mechanism estimates of the Butterworth filter are more realistic and less important than some types of filters. The Butterworth filter has significantly better amplitudes and transients [2]. Another main concern of this research is to use this designed filter in real and practical execution. All the simulation and results for the analyzing of different parameters have been analyzed and verified using FDA tool in MATLAB.

II. LITERATURE REVIEW

A lot of work has been done for digital filters used in signal processing. Fazlul Haque is committed to the enviable parameter extraction and uses FDA Tool [3] to improve the performance of the digital filter. Mohammad Saiful Islam, Shaila Shanjada and Mithun Banik are committed to using FDATool [4] to extract enhancements from digital filters. Yaduvir Sing, Sweta Tripathi and Manoj Pandey used LabVIEW [5] to analyze the digital IIR filter. Zhang Chengliang and Wang Aihong are engaged in the design research of IIR digital filter and simulation in MATLAB [6]. Ethan Elenberg et al. is engaged in the design of audio processing digital filters [7]. Uh. Daljit Singh Bajwa, Er. Karamjeet Singh and Navpreet Kaur Chahal dedicated to IIR digital filter design investigation paper [8]. Ekta Yadav and Rupali studied the design of digital IIR filters [9]. In the above research papers, all are based on analysis and design of digital IIR filter. Different types of design methods have been used for these analysis purposes, such as window function methods, pulse invariant methods, and some of them have used LabVIEW software to study. assist. In this article , different filter forms such as high pass filter, low pass filter, notch filter and band pass filter were used to test the performance of digital filter. The Butterworth IIR design method was used for this purpose. A tool was also developed by for filtering audio signals.

III. METHODOLOGY

In this paper, the actions of different digital filters and the responses have been studied and analyzed based on different parameters. Also, the filter output of a selected audio signal has been examined. MATLAB FDA (Filter design and analysis) tool has been used for analysis part tool.

A. Why IIR filter instead of FIR filter:

Mainly, there are two categories of digital filters. Infinite impulse response (IIR) and Finite impulse response (FIR). Filters which usually have an impulse response containing only a finite number of values are typically called finite impulse response (FIR) filters [10].

i. FIR Filter

A finite impulse response filter can be expressed by z-transform transfer function from input $X(z)$ to output $Y(z)$ as

$$H(z) = \frac{Y(z)}{X(z)} = \sum_{n=0}^{N-1} b_n z^{-n}$$

ii.. IIR Filter

Infinite impulse response filters can be expressed by rational form z-transform

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{n=0}^{N-1} b_n z^{-n}}{1 + \sum_{m=1}^{M-1} a_m z^{-m}}$$

FIR filters relate to the problem of filtering , which has conditions for the characteristics of linear phase within the passband of the filter. The design of the FIR filter focuses on a direct approximation of a response of a particular size, often adding the obligation that the phase response is linear [11]. On the other hand, a filter having an infinite dimensional impulse response is referred to as an infinite impulse response (IIR). In this paper, the IIR filter was applied to analyze other parameters. This is because the IIR filter has more efficient filtering

characteristics than the FIR filter. IIR filters are typically used in applications that do not require a linear phase response. In other words, the IIR filter usually has a non-linear phase response [12]. The IIR filter requires less memory as well as the FIR filter because it is faster than the FIR filter and requires fewer elements to do the filtering. Therefore, IIR filters have been implemented.

B. Prior knowledge to design a Digital Filter

For designing a filter to denoise an audio signal, we need to preprocess the audio file and analyze. In this project, we have tried to design a filter to remove unwanted noise from an audio file. We want to implement different types of filters such as Butterworth lowpass filter and others. The designed filter is implemented in MATLAB and used to filter the audio and compare to the unfiltered audio.

i. Basics of Human Voice

The air passes through the vocal cord and makes a sound, by controlling the vocal cords, the voices and the change of tone. The filtrate is applied to the spectrum of voice by affecting the cavities in the vocal cords (pharynx, oral, nasal). Change vocal efforts Changes both sound and frequency spectra. Even with the tone of the voice, it depends on the effort of the voices. The voices shouted are different from talking with casual voices. When recording, we can see that the peak of the acoustic signal is much higher than the RMS or the average level, mainly in the frequency range above 500 Hz. More specifically, in the frequency range of 2 kHz-4 kHz.

Using theoretical knowledge, it can be assumed that since the audio file is shown below in the figure:

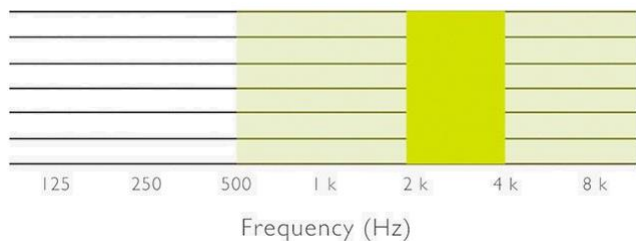
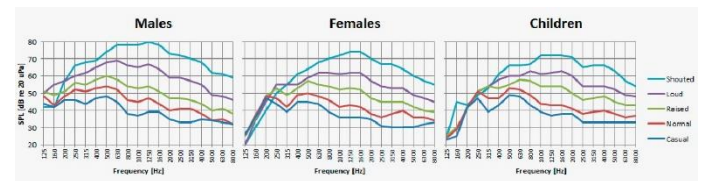


Fig. 1 Frequency Range of a Normal Human voice

ii. The spectrum of speech

Typically, the fundamental frequency of a complex voice - also known as pitch or f_0 - is between 100-120 Hz for men, but variations outside this range can occur. F_0 for women is about an octave higher. For children, f_0 is about 300 Hz.

In terms of frequency, normal vocal pitch, vowel energy usually decreases rapidly above about 1 kHz. Note, however, that the emphasis on the spectrum of speech shifts one to two octaves in higher frequencies as the voice rises. Also note that it is not possible to increase the sound level of a consonant to the same level as a vowel. In practice, this means that speech intelligibility does not increase when shouted, compared to applying normal voice, comparisons are shown below:



The important frequencies in non-tonal (Western) languages are illustrated by the diagram below. Here, the frequency band around 2 kHz is the most important frequency range regarding perceived intelligibility. Most consonants are found in this frequency band.

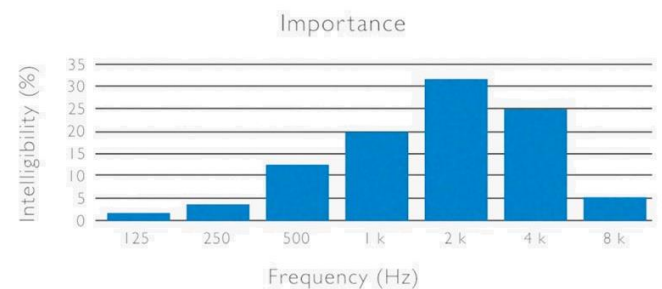


Fig. 2 Frequency Range and Intelligibility of Human Voice

The background noise affects the perceived understanding of audio signals. In this case, all the signs other than discourse themselves can be considered as noise. Therefore, in general noises are also caused digitally while sampling and quantization, in addition, if there are other people, noise will be produced. Television sounds and movies are very frequently related to the relationship between the level of dialogue and the level of background noise.

IV. SIMULATION & ANALYSIS:

To begin with the filter process, we need to access the noisy audio file in MATLAB and analyze, and choose the best possible parameter to filter the audio signal. The code for the following project is given in the appendix of the paper.

A. Performance of different filters at different Orders:

i. Low pass filter:

a. Magnitude Response:

From the amplitude response, IIR Butterworth can easily discover the basic properties of the filter. *Figure 3* shows. The filter only passes the low frequencies and rejects the highest frequencies. IIR Butterworth uses different orders, such as 400, 600, 800, 1000, and 1500, to find out which one can give a better result. It was later found that the output of the level 1500 was sharper. Pass the lowest frequency below the cutoff frequency.

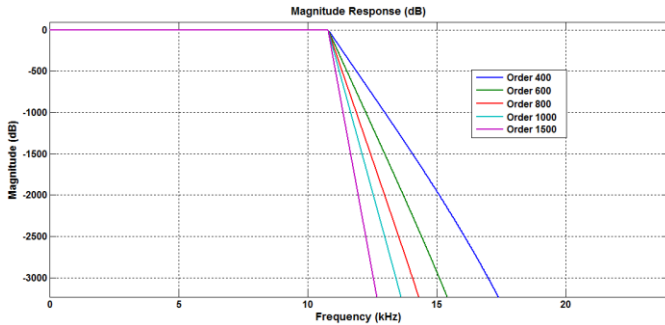


Fig. 3. Magnitude Response of low pass filter for different orders

b. Phase Response:

The phase response is one of the most important digital filter parameters of . It shows the relationship between the phases of the input and output signals. The phase response of the low-pass filter is combined with certain applications, such as time delay simulation, cascaded filter stages, especially for process control loops. The Butterworth IIR low-pass filter uses different orders (such as 400, 600, 800, 1000, and 1500 to) to evaluate the output shown in *Figure 4*.

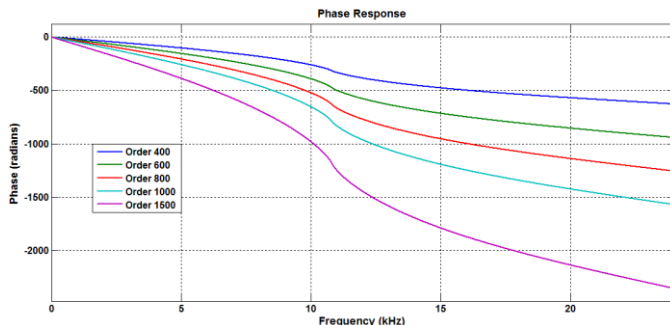


Fig. 4. Phase Response of low pass filter for different orders

ii. High pass filter:

a. Magnitude Response:

This is based on high pass filter and high pass filter which can transmit high frequency signal. Now, IIR Butterworth discusses high-pass filters in amplitude response and tries to find out under what situations a high-pass filter can give better results than . Here, *Figure 5* shows amplitude response of high-pass filters. Butterworth takes different orders like 400, 600, 800, 1000, 1500 and gives a different result. In FDA tool, the Butterworth IIR high-pass filter was used, and found that the response quality of 1500 orders of magnitude better than that of the other filters.

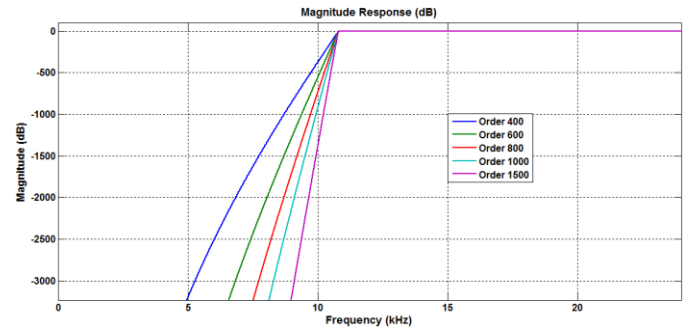


Fig. 5. Magnitude Response of high pass filter for different orders

b. Phase Response:

When phase feedback has been discussed, it can be said that it is part of signal processing. If the Butterworth filter IIR wants an effective phase response in a high pass filter, it measures a number on the order of -400, 600, 800, 1000 and 1500 shown in *Figure 6* contains inferior properties. From the high pass filter and the response of 1500 commands is stronger. *Figure 6*. Phase response of the high pass filter for different orders

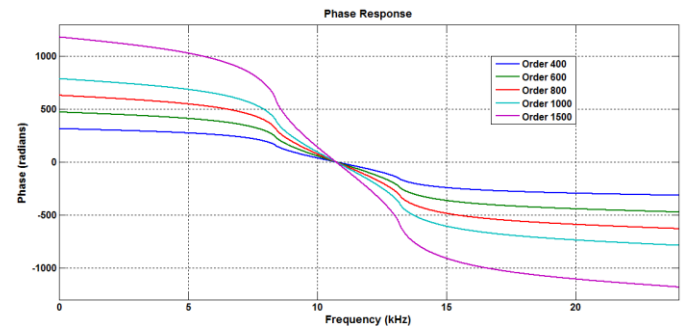


Fig. 6. Phase Response of high pass filter for different orders

iii. Band Pass filters:

a. Magnitude Response:

Magnitude response in IIR Butterworth band pass filter is used to find out the basic characteristics of digital filter. The best magnitude response in band pass filter will be found out. For this, different filter orders like- 400, 600, 800, 1000 and 1500 have been used which showed in the *Figure 7* and it is definitely high filter order delivered the better output. So, order 1500 is sharper and magnitude response takes the best output.

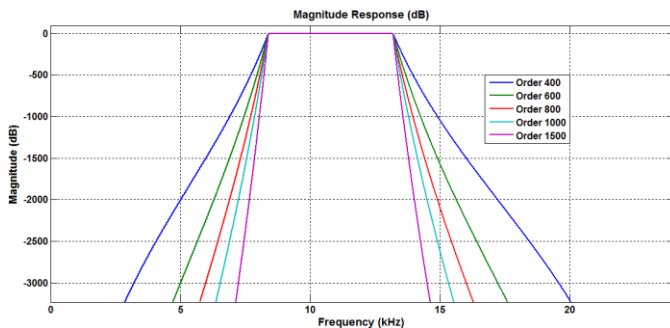


Fig. 7. Magnitude Response of Band Pass filter for different orders

b. Phase Response:

In a Butterworth IIR bandpass filter, the connection between the phase of the sine wave input and the output signal passing through any processor is called a phase response. Here in *Figure 8*, IIR Butterworth uses three orders of filters so that gets the best answer where the order of 1500 gives results in favor.

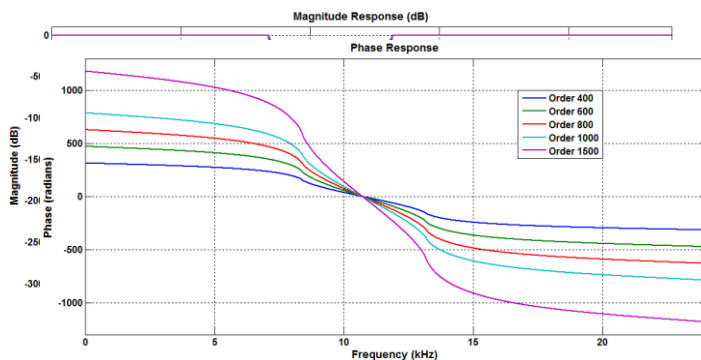


Fig. 8. Phase Response of Band Pass filter for different orders

iv. Band stop filter:

a. Magnitude Response:

The magnitude response of a band stop filter is a filter that passes most frequency response is unchanged. A general ideal band stop filter can display both positive and negative angular frequency. In IIR Butterworth band stop filter in the *Figure 9* takes different filter order, order 400, order 600, order 800, order 1000 where higher order 1000 gets more priority and gives accurate output.

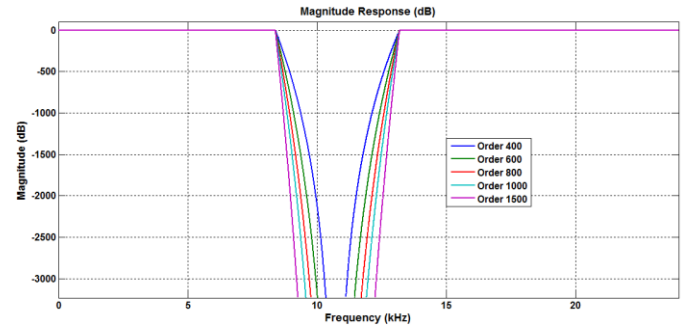


Fig. 9. Magnitude Response of Band Stop filter for different orders

b. Phase Response:

In IIR Butterworth band stop filter, phase response is the stage of the output with the input as reference. In *Figure 10*, IIR Butterworth method gets different filter order for taking favorable output. Higher filter order deployed much better output

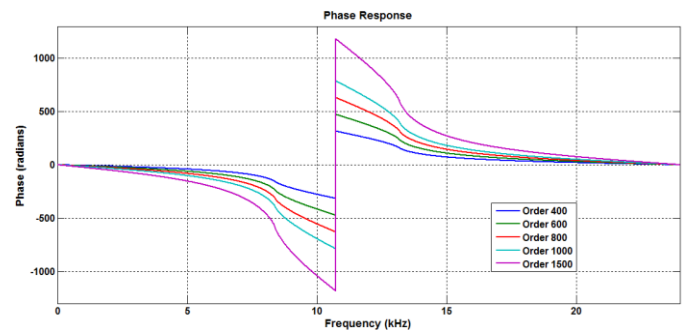


Fig. 10. Phase Response of Band Stop filter for different orders

B. Analysis of the provided Noisy Audio Signal:

i. Magnitude & Phase of the Noisy Signal

The audio file with noise that is provided is 4 seconds long and has a maximum amplitude of 0.9394 at 1.881 second and has a minimum amplitude of -1 at 1.036 second.

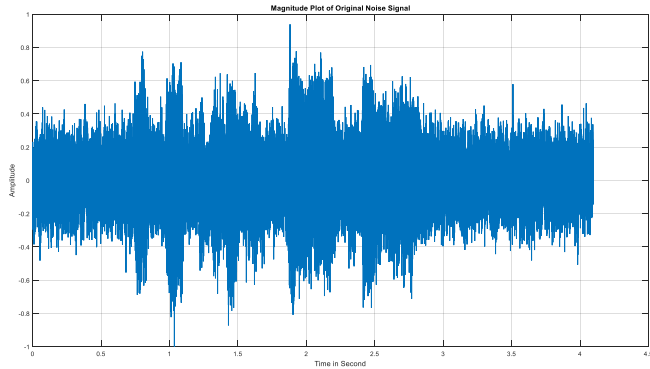


Fig. 11a. Magnitude vs Time plot for the Noisy Audio Signal

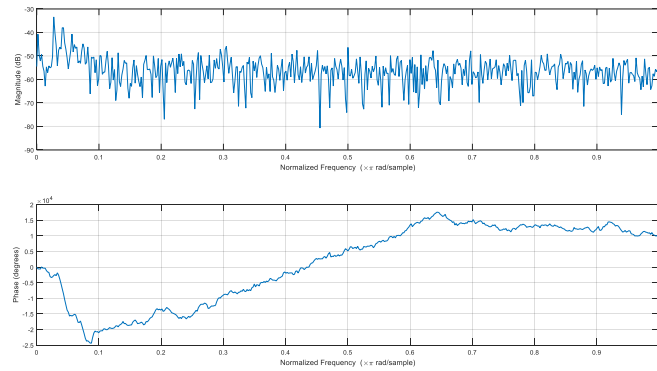


Fig. 11b. Magnitude (in dB) and Phase Plot against Normalized Frequency for the Noisy Signal

ii. Spectrogram of the Noisy Signal

The spectrogram is used to analyze where the frequencies are at most powerful, as can be seen below in the *Figure 12* their data available at frequencies below 1000Hz and also between 3500Hz to 2000Hz

It can be assumed that the majority of the useful information is between 3500Hz and 300Hz, but for further analysis we can transform the audio file into a frequency domain for better analysis.

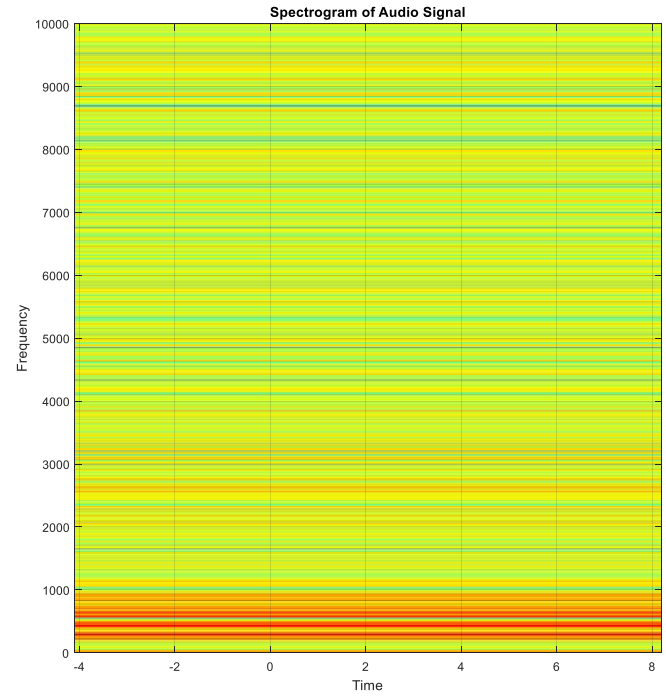


Fig. 12. Spectrogram for the Noisy Signal

iii. Fourier Transform of the Noisy Signal

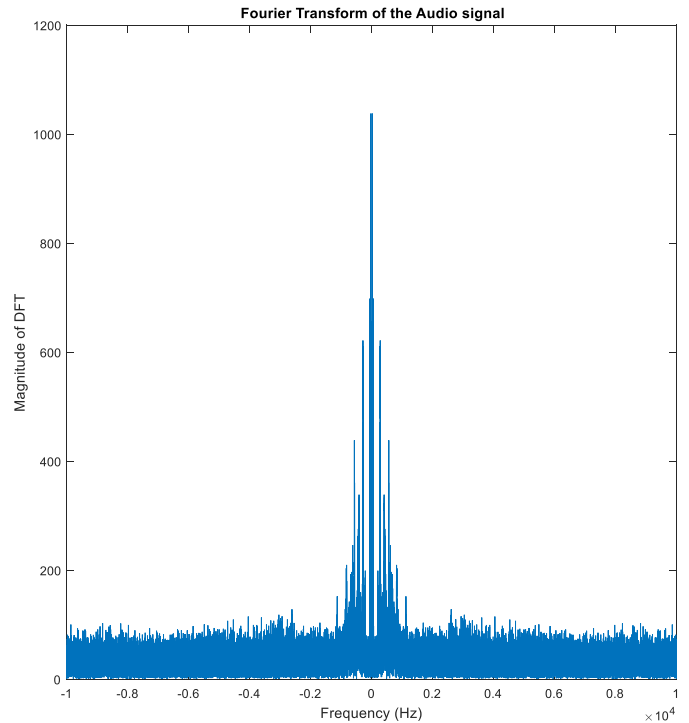


Fig. 13. Fourier Transform of the Noisy Signal

From *Figure-11a* the audio signal has been transformed into frequency-domain from time-domain in *Figure-13*, for noise reduction to recover a target signal from an input waveform is important in a number of fields. We have used a frequency spectrum to remove noise from the input waveform. Although it is difficult to distinguish a signal from the noise in the time domain, this task tends to become easier in the frequency domain. However, it is difficult to filter out noise that is similar to a signal, but we can at least deduct the signals at frequencies which are not in the range of the human voice frequency range which will minimize noise greatly.

Below is the magnitude response of the filtered audio signal in *Figure- 14a* and the Discrete Fourier Transform of the filtered audio signal as well in *Figure-14b*.

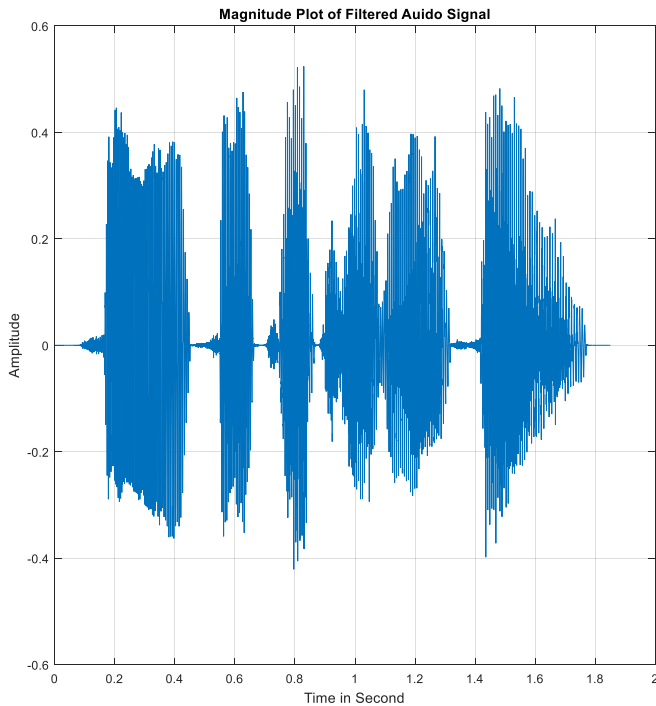


Fig. 14a. Magnitude vs Time Plot of the Filtered Audio Signal

In *Figure-14a* the Magnitude vs Time Plot of the filtered audio signal is shown, parts of the Amplitude is shrunk completely at different times. It is also noted that the original signal was 4 seconds long whereas the filtered audio signal is 1.83 seconds and the maximum amplitude of the original signal is 0.9394 and minimum amplitude at -1. In the filtered audio signal the maximum amplitude is 0.524 and the minimum amplitude is -0.421.

It can be concluded that implementing the Low-Pass Butterworth Filter has successfully reduced noise components from the original signal.

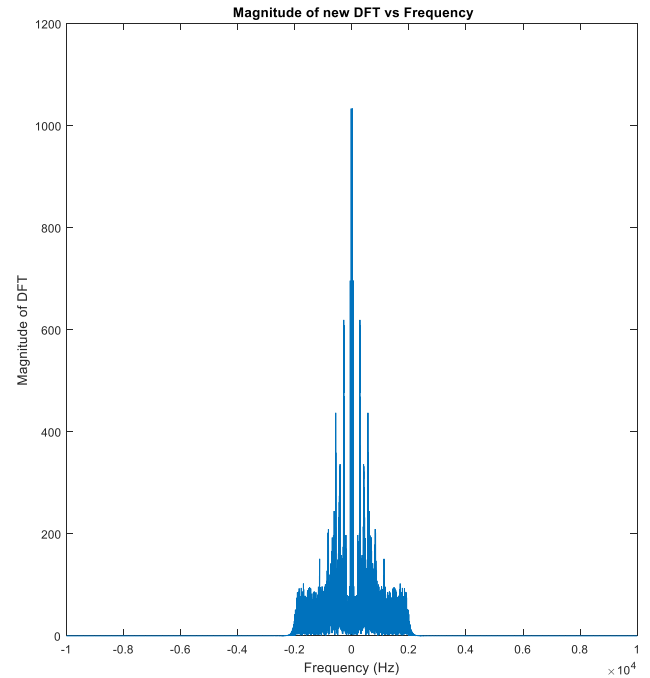


Fig. 14b. Discrete Fourier Transform of the Filtered Audio Signal

The Discrete Fourier Transform of the filtered audio signal is shown above in *Figure-14b*, the components at frequencies of below and above -0.2×10^4 Hz and 0.2×10^4 Hz are all cancelled out from the signal.

V. CONCLUSION:

In this paper, different design parameters have been considered for IIR Butterworth filter to analyze the filter performance. Four categories of filters like high pass, low pass, band stop and band pass filters have been used for IIR Butterworth filter. To analyze the performance, MATLAB FDA tool has been used. To simulate the result in FDA tool, different orders like order 400, 600, order 800, order 1000 and order 1500 have been considered. From the performance of different order, it has been analyzed that the better output can be found from higher order. That means, better result has been found from order 1500 comparing order 1000, 800, 600 and order 400. But higher order means the cost of implementing means more cost, the best performance of noise reduction of the signal has been achieved by using a Low-Pass Filter.

VI. APPENDIX:

Code for the following Project:

```
% Filter designed by Sowrav Chowdhury
% 183016008
% 12/5/2020

%{
Filter Design Project

Description:

In this project, we have tried to design a filter to remove
unwanted noise
from an audio file. We want to implement different types of
filters such as
Butterworth lowpass filter and others. The designed filter
is implemented in
MATLAB and used to filter the audio and compare to the
unfiltered audio.

Parts:
(1) Audio Frequency Analysis
(2) Filter Design
(3) Filter Implementation
(4) Report (attached pdf)

Included:
-Signal Processing Toolbox for MATLAB
-noise.wav (audio file that we will be filtering)

Outputs:

-Filtered audio file: filteredaudio.wav
%}

clc
clear all
close all

% 1. Audio Frequency Analysis
audioFileName = 'noise.wav';
[y,Fs] = audioread('noise.wav');
audio_DFT = fft(y);
n = length(y);

Y = ['The Sample rate of the signal is', num2str(Fs) 'Hz'];
disp(Y); %Displaying the value of Fs

%plot original signal in time domain
figure(1);
plot((1:n)/Fs, y);
title('Magnitude Plot of Original Noise Signal');
xlabel('Time in Second');
ylabel('Amplitude');
grid on;

figure(2);
specgram(y,length(y),Fs);
title('Spectrogram of Audio Signal');
grid on;

figure(3)
freqz(y, Fs);

% form a freq axis for plotting the DFT
freqAxis = linspace(-(Fs/2), (Fs/2), length(y));

% plot the magnitude DFT vs frequency
figure(4)
plot(freqAxis,fftshift(abs(audio_DFT)));
xlabel('Frequency (Hz)');
ylabel('Magnitude of DFT');
title('Fourier Transform of the Audio signal');

% 2. Filter Design
% Wp = end of the pass band, Ws = beginning of stop band
Wp = 1900; % 1.9 kHz
Ws = 2550; % 2.35 kHz
Rp = -1; % -1 db
Rs = -60; % -60 db
minAtten = 0.5;

% Solve for the nth order
args = (10.^(-Rs/10) - 1) / (10.^(-Rp/10) - 1);
N = (log10(args)) / (2*log10(Ws/Wp));
N_rounded = round(double(N))+1;
FqCutOff = Wp/(10.^(-Rp/10)-1)^(1/(2*N_rounded));

% Find Ha(s)
Ha_s =
20*log10(sqrt(1./(1+(freqAxis/FqCutOff).^(2*N_rounded))));

% 3. Filter Implementation

% first find Wn
Wn = FqCutOff/(Fs/2);
[b,a] = butter(N_rounded, Wn);
new_audio_DFT = fft(filter(b,a,y));

% plot the filtered signal:
figure (5)
plot(freqAxis,fftshift(abs(new_audio_DFT)));
xlabel('Frequency (Hz)');
ylabel('Magnitude of DFT');
title('Magnitude of new DFT vs Frequency');

%=====

% Prompt 1 for original audio file
[q,Gs] = audioread('prompt1.mp3');
sound(q,Gs);
pause(2);
% Play noise.wav
sound(y,Fs);
pause(12);

% Prompt 2 for filtered audio file
[q,Gs] = audioread('prompt2.mp3');
sound(q,Gs);
pause(2);
% play the filtered audio:
sound(filter(b,a,y),Fs);
% write new audio to a file
audiowrite('filteredaudio.wav',filter(b,a,y),Fs);

%plot original signal in time domain of filtered audio
signal
n2 = length(q);
figure (6);
plot((1:n2)/Gs, q);
title('Magnitude Plot of Filtered Audio Signal');
xlabel('Time in Second');
ylabel('Amplitude');
grid on;
```


VII. REFERENCES

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